Luca Drole

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Education

ETH Zürich — Switzerland
M.Sc. Biomedical Engineering (Biomedical Imaging) – 5.5/6 — *2023 – 2026*

Politecnico di Milano — Italy
B.Sc. Biomedical Engineering – 110/110 cum Laude — *2020 – 2023*

Experience

Computer Science PhD student — Jul 2026 – Present
TUM (Institute for Computational Imaging and AI in Medicine)

- Machine learning for quantitative MRI, reconstruction and image understanding. Supervised by J. Schnabel.

Master's Thesis – Vision Transformers for Medical Imaging — Oct 2025 – Present
ETH Zürich (Computer Vision Laboratory)

- Developing Vision Transformers backbones with context-aware downsampling for medical imaging

Teaching Assistant — Sept 2024 - Jan 2026
ETH Zürich (D-ITET, D-BAUG)

- Student support and course material development for "Programming for Engineers" and "Biomedical Imaging"

Visiting Research Student — Mar 2025 – Sept 2025
Martinos Center, Massachusetts General Hospital & Harvard Medical School — *Boston, MA*

- Developed contrast-agnostic MRI feature extraction backbone
- Applied deep representation learning to heterogeneous MR datasets, reducing brain-age prediction error by 20% against the baseline

Semester Project Student — Feb 2024 – Jun 2024
ETH Zürich (Cardiac Magnetic Resonance Group)

- Built CycleGAN-based framework for LGE-MRI to scar-mask translation (38 patients), achieving a median scar Dice of 0.88.
- Trained and benchmarked GAN architectures in PyTorch and TensorFlow

Research Intern (Amgen Scholar) — Jul 2022 – Sept 2022
ETH Zürich (Laboratory for Movement Biomechanics)

- Automated joint implant pose estimation from fluoroscopy videos and missing frame imputation, combining classical CV and DL approaches
- Presented results at the Amgen Scholars European Symposium (Cambridge), runner-up best-poster prize

Publications & Projects

- ***Towards a Lightweight 2D U-Net for Accurate Semantic Segmentation of Kidney Tumors in Abdominal CT Images***, IEEE EUROCON 2023. First-author conference paper on efficient semantic segmentation architectures for abdominal CT imaging.
- ***Medical Image Generation on a Budget*** (Course Project, GitHub). Developed and evaluated diffusion-based MRI synthesis under strict computational constraints; implemented LoRA-based adaptation strategies and benchmarking pipelines (FID, classifier AUC).

Honors

- ETH Zürich Excellence Scholarship (CHF 48,000) – Global **Top 60** incoming Master's students
- Nova 111 Italy (2025) – Top 10 students nationwide in Engineering & Technology

Skills

Programming: Python, MATLAB
ML: PyTorch, TensorFlow, Vision Transformers, GANs, CNNs, UNet, Diffusion Models
Medical Imaging: MRI, CT, Segmentation, Multimodal Analysis
Tools: Git, Docker, Slurm, Linux, LaTeX
Languages: Italian (native), English (C2), German (A2)